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BSIT 3-2

Application - Development

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A full-stack application consists of multiple layers that work together to deliver a complete software system. These layers include the frontend, backend, database, and API. Each component has a specific role in handling user interaction, processing data, storing information, and enabling communication between systems.

The frontend is the user interface where users interact with the system. It can be a web interface built with HTML, CSS, and JavaScript, or a mobile interface built using Kivy. The frontend is responsible for collecting user input and sending requests to the backend.

The backend is the server-side part of the application that processes requests from the frontend. In this architecture, the backend is built using Django, which handles application logic, authentication, and communication with the database.

The database stores all persistent data used by the application. SQLite is commonly used for local development because it is lightweight and easy to configure, while PostgreSQL is used for cloud or production environments due to its scalability and advanced features.

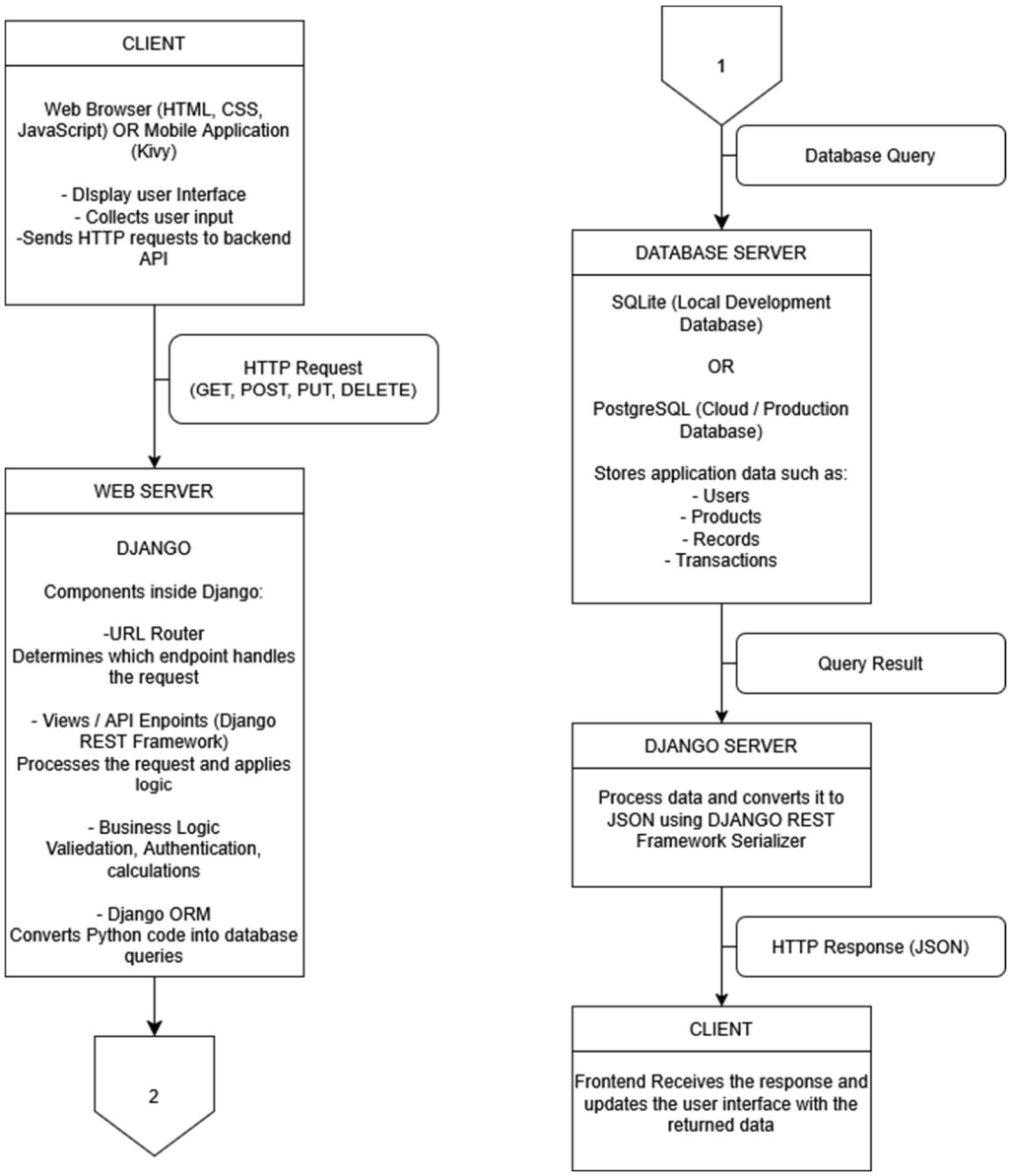
The API acts as the communication layer between the frontend and backend. Using Django REST Framework, the backend exposes RESTful endpoints that allow the frontend to send and receive data in a structured format such as JSON.

When a user interacts with the application, the request begins at the client side through a web browser or mobile app. The frontend sends an HTTP request to the Django web server through the REST API. The Django backend processes the request, applies the required business logic, and retrieves or stores data in the database. The database then returns the requested information to the backend. Finally, the backend sends a response, usually in JSON format, back to the frontend, where the data is displayed to the user.

The system uses Django as the backend web framework to manage server logic and handle requests. Kivy is used to build the mobile frontend interface using Python. SQLite serves as the local development database, while PostgreSQL is used as the cloud or production database for larger-scale applications. Django REST Framework provides the tools needed to build REST APIs that allow communication between the frontend and backend.

The frontend communicates with the backend by sending HTTP requests to the REST API endpoints provided by Django. The backend processes these requests and interacts with the database using Django's Object Relational Mapping (ORM) system. The database

stores and retrieves application data, which is then returned to the backend. After processing, the backend sends the results back to the frontend as a response, allowing the user interface to display the updated information.

A full-stack architecture integrates the frontend, backend, API, and database to create a complete application. Each component has a specific responsibility but works together to process user requests and deliver responses. By using Django, Kivy, SQLite, PostgreSQL, and Django REST Framework, developers can build scalable applications that support both web and mobile platforms.

**Architecture Diagram**